Who Decides? A System One Model in the Decision Seat of Clinical Agents

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Abstract

Clinical agents now read records, order tests and write notes, and every step they take is a decision generated by a large language model (LLM) after thousands of tokens of reasoning. System One decision models make a decision in a single forward pass: they read a state and a set of options and return a probability for each. We ask whether such a model can take the decision seat inside real clinical agent loops. We hold everything else fixed: the backbone (Clef-27B is post-trained from Qwen3.8-27B, our LLM), the benchmarks' own loops, tools and graders, and the hardware. On PhysicianBench, 100 consultation tasks run against an electronic health record of real de-identified patients, a System One model that chooses among five turns proposed by the LLM matches the LLM: 80.2% of graded checkpoints versus 81.3%, with no significant difference in task success. It is the more decisive actor. When a required order is among the proposals, it places it in 100% of cases; the LLM's own sample does so in 38%. Asked instead to steer the agent, choosing which tool to use next, the same model stops gathering evidence too early and passes only 44.3% of checkpoints, while the LLM making that same restricted choice loses almost nothing. Step-level analysis on MIMIC-IV admissions shows what limits selection: the LLM's samples are usually the same decision in other words. Our results point to a clean division of labour. The LLM proposes and gathers; a System One model decides among concrete actions.

1 Introduction

A physician's agent takes dozens of steps per task: look up the problem list, pull the labs, read last month's note, order the imaging, write the assessment. Each step is a choice among a small set of concrete actions, and in today's systems an LLM makes every one of them by generating text (Liu et al., 2026; Lu et al., 2026; Yao et al., 2023). Figure 1 follows one such task.

System One decision models offer a different way to choose (Almeida, 2026; Cloudflare, 2026). They take a state and a typed question with named options and return a probability for every option in one forward pass, with no generated text and nothing to parse. They are fast, cheap and well suited to choosing among options, the operation an agent performs at every step. Whether they can make an agent's decisions is open. So far they have been studied on single-shot tasks and as judges (Li et al., 2026b), and in general-purpose agent loops their actions often go wrong even when they read the state correctly (Li et al., 2026a). No study has placed one in the decision seat of a clinical agent.

We do so under the strictest comparison we can construct. Clef-27B is post-trained from Qwen3.8-27B, which serves as our LLM, so the two decision makers share a backbone and differ in how they read out a decision. We run two clinical benchmarks in their own harnesses, with their own tools, simulators and graders, and change only the component that picks the next turn (Figure 2). The System One model either chooses among complete turns proposed by the LLM (S1-SELECT) or chooses only the kind of action, leaving the LLM to write it (S1-TOOL).

Findings.

- **A System One model can make the agent's decisions.** On PhysicianBench, S1-SELECT passes 80.2% of graded checkpoints against the LLM's 81.3% (Δ −4.0 [−13.0, +5.0] points in task success), with no extra generated text from the decider (§4.1).
- **It acts when action is required.** When one of the proposed turns places an order the task requires, the System One model takes it in 100%

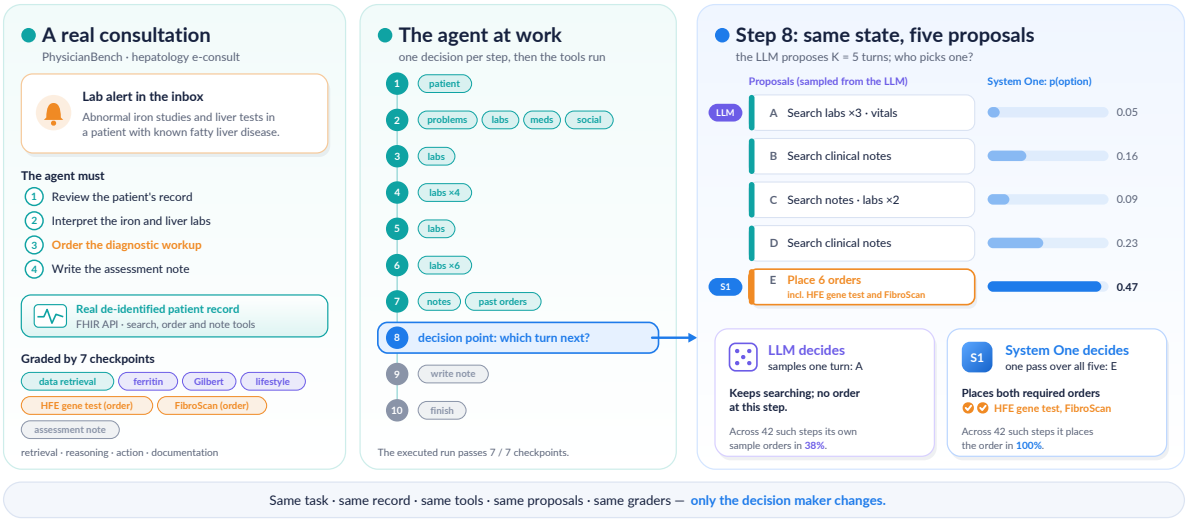


Figure 1: **One real consultation, one decision, two deciders.** A PhysicianBench task (instruction paraphrased): a hepatology e-consult on abnormal iron and liver tests, in which the agent must review the record, interpret the labs, order the workup and write a note, graded by seven checkpoints. The agent works step by step through the patient's record. At step 8 the LLM has proposed five next turns. Its own sample (A) keeps searching labs. The System One model reads the same state and all five proposals in one forward pass and puts the most probability on E, which places both orders the graders require. Across all 42 order steps where a proposal places a required order, the LLM's own sample orders in 38% and the System One model in 100%. The figure shows tool types, probabilities and outcomes only, never patient data.

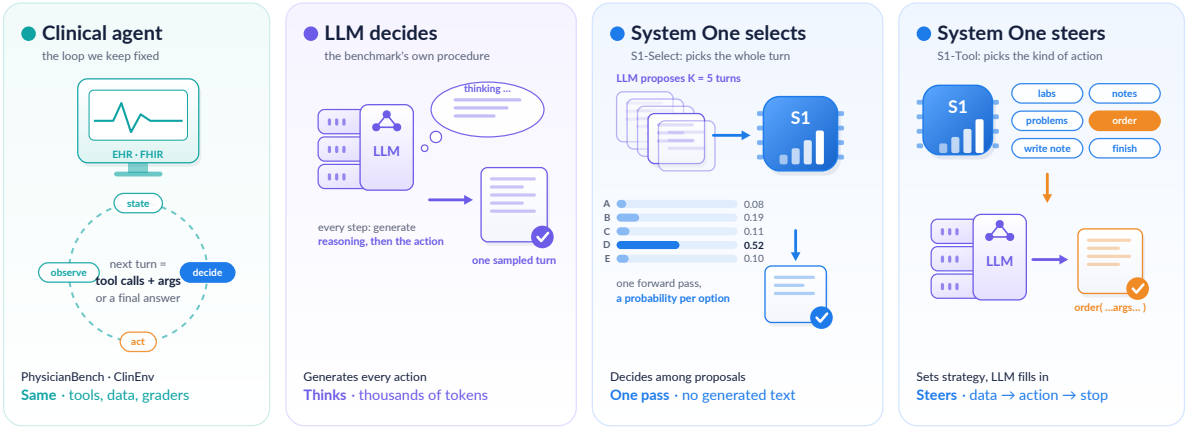


Figure 2: **Three ways to make the next decision in the same clinical agent loop.** The loop, tools, records and graders stay fixed. LLM: the model samples its next turn after reasoning. S1-SELECT: the LLM proposes five complete turns and a System One model picks one in a single forward pass. S1-TOOL: the System One model picks the kind of action from the full tool menu and the LLM writes the call.

of steps, against 38% for the LLM's own sample; when no required order is on offer, it orders far less often (54%). Action-execution checkpoints move by +5.1 [−2.4, +13.9] points (§4.2).

- **It should not steer alone.** Choosing the tool for the LLM, the same model cuts data retrieval from 31.2 to 14.2 calls per task and writes the note 9.2 times; checkpoint success falls to 44.3%. A control with the same one-tool restriction but the LLM choosing stays within noise of the LLM (−0.5 [−4.7, +3.9] points), so the loss lies in the System One model's choices (§4.3).

- **Other System One models behave the same way.** On one fixed exam of logged states and proposals, pplx-decider-v1.2 and d3 take every required order on offer, as Clef does, while the LLM's own sample takes half; online, pplx-decider-v1.2 again matches the LLM when se-

lecting and falls behind when steering (§4.4; preliminary).

- **Selection is bounded by what the proposer samples.** On MIMIC-IV admissions, 80% of the LLM's five-sample candidate sets contain a single clinical decision, and both deciders follow the proposer's hypothesis even when the gold decision is added (§4.5).

2 Putting a System One model in the decision seat

An agent loop alternates between a state s_t, the task plus every action and result so far, and the next turn a_t: a set of tool calls with arguments, or a final answer. The benchmark samples $a_t \sim \pi(\cdot \mid s_t)$ from the LLM. We replace only this step.

LLM (reference). One sample from π with thinking on and the model's default sampling, exactly as the benchmark prescribes.

S1-SELECT: the System One model chooses the turn. The LLM samples $K{=}5$ turns $a_t^{(1)}, \ldots, a_t^{(5)}$ i.i.d. from π in one request. Exact duplicates are merged. The System One model reads the state and the distinct turns, rendered as tool calls with full arguments, and returns a probability for each; the most probable turn is executed. It never sees the LLM's reasoning, so it judges actions, not rationales. A uniform pick among K i.i.d. samples has the same distribution as LLM, so any difference between the two comes from *which* sample is chosen.

S1-TOOL: the System One model chooses the kind of action. The System One model reads the state and the full tool menu (each tool's name and description, plus FINISH) and picks one. The LLM then writes the turn with every tool still in its prompt but its calls constrained to the chosen tool. The System One model now sets the agent's strategy and timing (when to gather, when to order, when to document, when to stop) and the LLM supplies the content.

LLM-TOOL: a control for S1-TOOL. S1-TOOL also restricts each turn to one kind of tool, whereas the LLM usually batches several lookups. LLM-TOOL applies the same restriction with the LLM choosing: it samples its turn as usual and keeps only the calls of the turn's first tool.

Fair inputs. Options are named by letters shuffled with a seed per task, run and step. The System One model reads a deterministic 32k-token view of the state (oldest tool results shortened first); the LLM reads up to 128k tokens. If the System One model fails, the first sample is executed; this happened 0 times in the S1-SELECT run.

3 Setup

PhysicianBench (Liu et al., 2026) adapts 100 real outpatient consultations from 21 specialties into long-horizon tasks on a FHIR server with the records of 108 de-identified patients (47k conditions, 114k observations, 35k procedures, 13k medication requests, 2.3k clinical notes). Each task has 5–9 checkpoints (670 in total) of four types: data retrieval (DR), clinical reasoning (CR), action execution (AE: orders, prescriptions, referrals placed in the record and verified by code) and documentation (Doc). A task passes only if all of its checkpoints pass. We use the released harness: a fresh server and database per task, the MiniAgent loop with parallel tool calls and at most 100 steps, and the task's pytest graders.

ClinEnv (Lu et al., 2026) turns MIMIC-IV admissions (Johnson et al., 2023) into staged episodes. At each stage the agent questions simulated patient, nurse, lab and history agents, then submits medications, procedures and diagnoses, scored against the recorded decisions with ontology-aware matching. Every submission can be checked against gold, which makes ClinEnv our instrument for step-level analysis. We use the authors' 392 released admissions scored by all seven models in their paper.

Models. The LLM is Qwen3.8-27B (Qwen Team, 2026) in bf16 with 128k context and thinking on, served by vLLM (Kwon et al., 2023). The System One model is Clef-27B (Cloudflare, 2026) with its native joint head. The PhysicianBench judge is the same Qwen model with thinking off, shared by all arms. Everything runs on one node with four H100 GPUs; no record leaves it (§6).

Statistics. Arms are compared on the tasks both completed. We report differences with 95% bootstrap intervals over tasks (10,000 resamples) and exact McNemar tests for task success.

	Task pass@1		All checkpoints		Checkpoints by type (%)					
Decision maker	%	Δ [95% CI]	%	Δ [95% CI]	DR	CR	AE	Doc	Calls /task	Tokens /task
LLM	**41.0**	*ref.*	**81.3**	*ref.*	83.2	**84.1**	75.0	**70.7**	36.3	29k
S1-SELECT	37.0	−4.0 [−13.0, +5.0]	80.2	−1.1 [−4.7, +2.6]	**85.0**	82.8	**80.1**	65.7	31.2	119k
S1-TOOL	7.0	−34.0 [−45.0, −23.0]	44.3	−37.0 [−44.5, −29.3]	44.7	47.5	52.3	26.3	25.7	18k

Table 1: **PhysicianBench, run 1, 100 tasks completed by all three decision makers.** Δ: difference from LLM in points with a 95% task bootstrap interval. Checkpoint types: data retrieval (DR), clinical reasoning (CR), action execution (AE), documentation (Doc). Calls: tool calls per task. Tokens: generated tokens per task, including all five proposals for S1-SELECT. Bold: best per column.

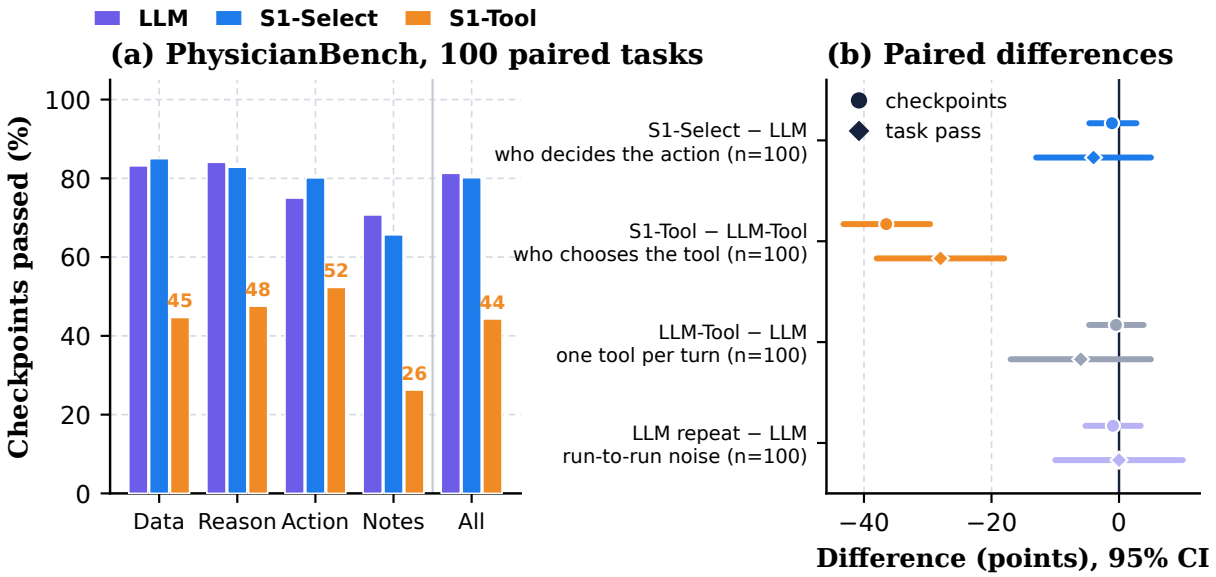


Figure 3: **A System One model matches the LLM when it chooses among proposals, and falls behind when it steers.** (a) Checkpoint success by type on the paired tasks. (b) Paired differences with 95% intervals, each on the tasks both arms completed: the System One model against the LLM; who chooses the tool under the one-tool restriction; and the restriction itself.

4 Results

4.1 Choosing among proposals, the System One model matches the LLM

Table 1 and Figure 3 give the headline. S1-SELECT passes 80.2% of checkpoints against 81.3% for the LLM (Δ −4.0 [−13.0, +5.0] points in task success, McNemar $p = 0.52$). Over all 100 tasks both have completed, the difference stays within a point (−1.1 [−4.7, +2.7] on checkpoints). It is not a wash, though: the two deciders behave differently. The System One model departs from the LLM's own first sample at 79% of steps with a real choice and reaches the same quality with fewer calls (31.2 versus 36.3 per task). Its profile shifts toward action: action execution +5.1 [−2.4, +13.9] and data retrieval +1.8 [−4.0, +7.8] points, documentation −5.1 [−13.1, +3.0].

4.2 It acts when action is required

Why does action execution improve? Every step of the S1-SELECT run logs the five proposals and the System One model's probabilities, so each order step can be scored against the orders the task's graders require (Figure 4). Over 83 order steps in 53 tasks, two kinds of step stand out. When one of the proposals places a required order, the System One model chooses an ordering turn 100% of the time, although only 39% of proposals order and the LLM's own first sample orders in 38% of these steps. When no proposal places a required order, it orders in 54% of steps. On the 43 steps that mix right and other proposals it picks a right one 91% of the time (random 35%, first sample 37%, candidate AUROC 0.95). The LLM often proposes the right order and then, sampling, fails to place it; the System One model recognises it.

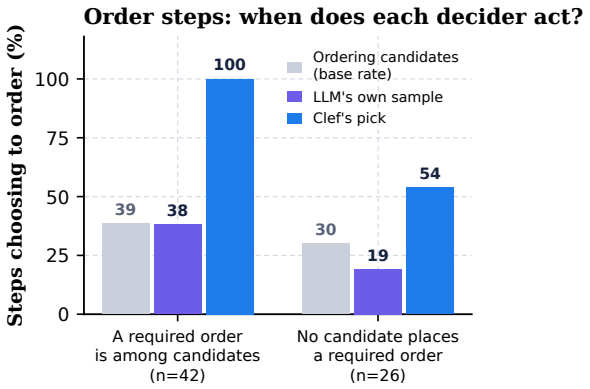


Figure 4: **The System One model places required orders.** Order steps of the S1-SELECT run whose proposals mix ordering and other turns. Orders the graders do not require are not wrong, only uncredited.

4.3 Steering alone, it stops gathering data

Given the tool menu instead of proposals, the same model fails. S1-TOOL passes 7.0% of tasks and 44.3% of checkpoints (Δ −34.0 [−45.0, −23.0] points in task success, $p < 0.001$), and its data-retrieval checkpoints collapse first. Figure 5 shows why: it retrieves 14.2 times per task against the LLM's 31.2, and it writes the note 9.2 times, rewriting it again and again instead of finishing. The LLM disagrees with its tool choice often enough that 56% of steps must be regenerated with the tool named.

The control settles where the loss comes from. Restricting the LLM to one tool per turn costs nothing measurable: LLM-TOOL differs from LLM by −0.5 [−4.7, +3.9] points on checkpoints over 100 tasks. Handing the same restricted choice to the System One model costs −36.5 [−43.2, −29.6] points (S1-TOOL versus LLM-TOOL, 100 tasks; task success 7.0% versus 35.0%, $p < 0.001$). The tools it picks explain the gap (Table 2): it chooses to write the note at 44% of steps, where the LLM does so at 7%, and it spends 42% of its steps on lookups against the LLM's 81%. Deciding *when* the evidence is sufficient is exactly the judgment the System One model lacks.

4.4 Other System One models in the same seat

Is this Clef's behaviour, or how System One models behave as deciders? We put five other open System One models in the same seat: pplx-decider-v1.2, d3, linne-oak-1.0, amos-1-large and decider-31b. Each runs its authors' own serving code at a pinned revision, and nothing else changes: the same rendering, question wording and 32k fitting rule (Appendix A).

Tool	Clef (S1-TOOL)	LLM (LLM-TOOL)
Write note	44	7
Search labs	17	31
Search notes	10	20
Search problems	4	9
Place order	7	4
Search med. orders	3	7
Demographics	5	5
Finish	4	4

Table 2: **Which tool each decider picks** under the same one-tool-per-turn restriction, as a share of steps (%), on the tasks both completed.

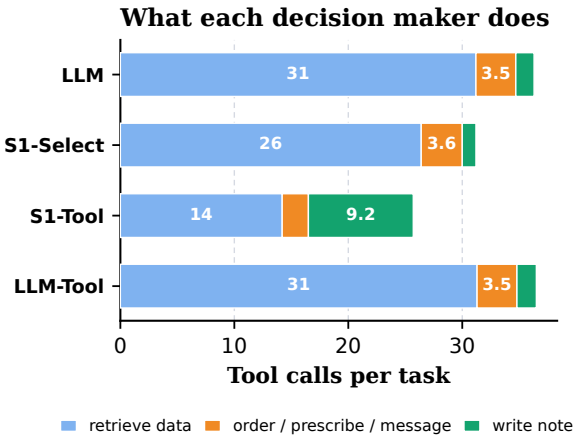


Figure 5: **Tool calls per task by kind.** Steering, the System One model under-retrieves and over-documents. LLM-TOOL: the LLM under the same one-tool-per-turn restriction, on the tasks it shares with S1-TOOL.

The same exam for every decider. The LLM runs every task once and logs, at each step, its five candidate turns and the full state. Each step with at least two distinct candidates becomes a question, so far 1466 questions from 100 tasks. Every decider answers byte-identical requests: the same state, the same options and the same option letters. The comparison is restricted to the 75 questions that all of them have answered. On order steps the task graders give an answer key (13 questions that separate right from other options). Table 3 reports the result. On order steps, all three deciders pick a right option far more often than the LLM's own sample (Clef 92%, pplx-decider-v1.2 100%, d3 92%, against 54% for the LLM and 45% at random). When a proposal places a required order, every one of them takes it (100%, 100% and 100%; the LLM 50%). The bias toward action is shared

	Order steps (answer key)			All questions		
Decider	Right	Orders, required	Orders, none req.	Agrees w/ LLM	Act	Note
Random	45	–	–	25	–	–
LLM (own sample)	54	50	0	100	–	–
LLM (most sampled)	54	–	–	–	–	–
Clef-27B	92	100	0	31	17	–
pplx-decider-v1.2	100	100	100	31	19	–
d3	92	100	0	25	17	–
linne-oak-1.0	92	100	0	23	17	–

Table 3: **The same exam for every decider** (PhysicianBench, the LLM's own logged states and candidates; 75 questions answered by every decider, 13 of them with an answer key). Right: picks a required order on order steps where options differ. Orders, required / none req.: picks an ordering option when one places a required order, or when none does. Agrees w/ LLM: same pick as the LLM's own sample. Act, Note: share of all picks that place orders or write the note. Preliminary: the exam and the remaining deciders are still running.

	S1-SELECT vs LLM		S1-TOOL vs LLM-TOOL	
Decider	Tasks	Δ checkpoints	Tasks	Δ checkpoints
Clef-27B	100	−1.1 [−4.7, +2.7]	100	−36.5 [−43.2, −29.6]
pplx-decider-v1.2	76	+1.1 [−5.0, +7.2]	49	−24.7 [−34.7, −14.8]

Table 4: **Whole tasks with another decider in Clef's seat.** Paired checkpoint differences with 95% intervals, each on the tasks both arms have completed.

too: when no proposal places a required order, they still order in 0%, 100% and 0% of steps, the LLM in 0%. Across all questions they agree with the LLM's own sample about as often as a random pick would (31%, 31% and 25%; random 25%), and with each other more often (pplx-decider-v1.2 and d3 on –% of questions). Taking the required action is how System One deciders behave, not a quirk of Clef.

Whole tasks. We also rerun the online arms with pplx-decider-v1.2 in Clef's seat (Table 4). Choosing among proposals, it again stays level with the LLM (+1.1 [−5.0, +7.2] checkpoint points over 76 tasks). Steering, it again falls far behind the same restriction with the LLM choosing (−24.7 [−34.7, −14.8] points over 49 tasks). Both online findings hold for a second System One model.

4.5 Selection is bounded by what the proposer samples

A selector can only add value where the proposals differ. ClinEnv lets us check every proposal against gold. In 24 admissions run with S1-SELECT, the five proposals are distinct strings at 100% of submit steps, because each submission carries a free-text

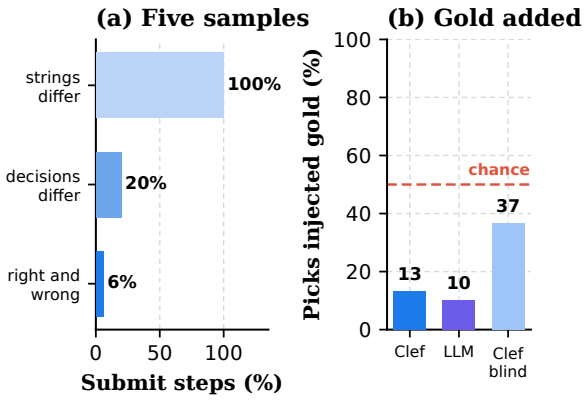


Figure 6: **ClinEnv submit steps.** (a) How often the LLM's five proposals differ as text, as clinical decisions, and in correctness. (b) With one gold decision added to two-way sets, both deciders follow the proposer.

rationale. Mapped to diagnosis and drug codes with ClinEnv's own matcher, 80% of these sets hold a single clinical decision, and only 6% contain both a right and a wrong one (Figure 6a). On PhysicianBench, where a turn bundles several calls, proposals differ far more often (89% of steps).

To probe discrimination directly we replay the states of an LLM run exactly, render every proposal as action plus canonical name, and where all are wrong, replace one by a gold decision rendered the same way. Every set then holds one right answer among two. Both deciders pick the proposer's option: the System One model chooses gold in 13% of 30 sets and the LLM in 10% (Figure 6b). Shown only the options, the System One model chooses gold more often (37%). The state is the LLM's own trajectory, whose questions and lab orders gather evidence for its working hypothesis; a decider that reads it follows that hypothesis. Step-level gold injection inside an agent's trajectory therefore measures agreement with the trajectory more than clinical discrimination, a pitfall for any step-wise evaluation of agents.

5 Related work

Choosing among samples. Majority voting over reasoning paths (Wang et al., 2023), trained verifiers (Cobbe et al., 2021), process reward models (Lightman et al., 2024) and LLM judges (Zheng et al., 2023) all pick among sampled outputs. S1-SELECT applies best-of-K selection at every step of an agent loop, with a decision model as the selector.

System One decision models. JEV introduced the typed-question interface (Almeida, 2026); Clef is an open 27B model with a joint head over a Qwen3.8-27B backbone (Cloudflare, 2026). As judges they approach frontier LLMs on verdicts that can be read off the input at a small fraction of the cost (Li et al., 2026b); as next-action selectors in general agents, most of their wrong actions follow a correct reading of the state (Li et al., 2026a). We bring them into clinical agents and separate choosing an action from steering.

Clinical agent benchmarks. PhysicianBench grades long-horizon physician tasks in a live record system by code and rubric (Liu et al., 2026); ClinEnv evaluates staged inpatient decisions against MIMIC-IV (Lu et al., 2026; Johnson et al., 2023). Both have been run only with generative LLM agents.

6 Discussion

A division of labour. The LLM is good at what the System One model cannot do: deciding what to look up, how much evidence is enough, and writing the note. The System One model is good at what the LLM does unreliably: committing to the right concrete action once it is on the table. S1-SELECT combines the two and loses nothing in checkpoint quality. S1-TOOL hands the System One model the part it is worst at.

Cost. The System One decision itself is one forward pass, about 4.7 s per step on one GPU, and adds no generated tokens. The cost of S1-SELECT lies in the five proposals (119k versus 29k generated tokens per task). Cheaper proposers, fewer proposals when they agree, or proposals targeted at decision steps would reduce it; the System One model never needs to reason in text.

What would make selection stronger. Proposals that differ in substance, not wording: when the five samples are one decision, no selector can help (§4.5). Diversity-seeking proposal schemes are the natural next step.

Limitations

We study one LLM. Our main System One model, Clef-27B, shares its backbone; the five other System One models (§4.4) are evaluated on a growing exam and, for one of them, online, so their results are preliminary. The PhysicianBench numbers come from one run per arm, and the LLM-TOOL control covers the 100 tasks it shares with S1-TOOL; repeated runs (pass@3) are in progress. Our judge is the local LLM rather than the leaderboard's, so absolute scores are not comparable with the leaderboard, while every arm shares the judge. The System One model reads a 32k-token view of states the LLM reads in full. ClinEnv enters as a step-level instrument; its online runs were paused to complete PhysicianBench.

Ethics and data

PhysicianBench's records are real de-identified patients under a data use agreement, and ClinEnv derives from MIMIC-IV, which forbids sending records to services without verified zero retention. All models run locally, a socket guard blocks any non-local connection from agent and grader processes, and trajectories and notes stay on access-controlled storage outside version control. We report aggregates only. ClinEnv's code has no license file at the version used; we use it for research reproduction.

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A Decision interface

Questions. The System One model receives the state and one Choice question. For S1-SELECT: *"The state shows a clinical task assigned to an AI agent that works in an electronic health record through FHIR tools, followed by the agent's actions and their results so far. Each option is a candidate next action for the agent. Choose the candidate that best advances completing the task correctly, completely and safely."* For S1-TOOL the second and third sentences read: *"Each option is a tool the agent can use for its next action, or finishing with a final answer. Choose what the agent should do next to complete the task correctly, completely and safely."* Options are the rendered turns (S1-SELECT) or each tool's name with the first paragraph of its description, plus FINISH (S1-TOOL). ClinEnv uses the same template with a description of its staged, conversational setting.

State. The state is the task instruction followed by every step: the action, rendered as tool calls with full arguments, and its results as the LLM saw them. It is fitted into the System One model's 32,768-token context deterministically: tool results are kept in full, then shortened oldest-first to 800 characters, then omitted oldest-first, each by the smallest number that fits. In the S1-SELECT run 44% of decisions saw shortened results and 1% saw omitted ones.

Other System One models. pplx-decider-v1.2 (`cce0f3f8`), d3 v3.1.0 (`cbff3a3c`), linne-oak-1.0 (`20da7bf3`), amos-1-large (`d67e051c`) and decider-31b (`e12ddf20`) each run their authors' own serving code on one H100, locally and offline. pplx-decider-v1.2 ships a library only, which we call on its reference path. decider-31b is NVFP4, which an H100 runs through vLLM's weight-only FP4 path; its authors used Blackwell GPUs. Every model receives the state and question built for Clef, with the same fitting rule, counted in its own tokenizer.

The LLM. Qwen3.8-27B with thinking on, temperature 1.0, top-p 0.95, top-k 20, at most 16,384 completion tokens per call. Two identical replicas serve all arms (one with tensor parallelism over two GPUs); each task is routed whole to the less loaded replica, independent of the arm.

B Additional results

Table 5 lists every paired comparison with intervals; Table 6 describes the decisions. Episodes that ended without a final answer (step limit, repeated-call abort or context overflow) number 2 for LLM, 1 for S1-SELECT and 16 for S1-TOOL on the paired tasks.

C Step-level evaluation on ClinEnv

Items. A submit step is a step whose proposals are all submissions of one type. Each proposal is scored against the stage's gold decisions of that

Comparison	Tasks	Pass@1	Δ pass@1 [95% CI]	p	Δ checkpoints [95% CI]	Δ DR	Δ CR	Δ AE	Δ Doc
S1-SELECT vs LLM	100	37.0 / 41.0	−4.0 [−13.0, +5.0]	0.52	−1.1 [−4.7, +2.7]	+1.8	−1.2	+5.1	−5.1
S1-TOOL vs LLM-TOOL	100	7.0 / 35.0	−28.0 [−38.0, −18.0]	<0.001	−36.5 [−43.2, −29.6]	−39.3	−36.9	−27.4	−38.4
LLM-TOOL vs LLM	100	35.0 / 41.0	−6.0 [−17.0, +5.0]	0.38	−0.5 [−4.7, +3.9]	+0.8	+0.4	+4.7	−6.1

Table 5: Paired comparisons, each on the tasks both arms completed (run 1). Δ in points; per-type columns give the point difference in checkpoint success.

	Steps /task	Distinct strings (%)	Distinct actions (%)	Mean max p	Norm. entropy	Departs from LLM sample (%)	Decide (s)
S1-SELECT	11.8	99	89	0.382	0.886	79	4.7

Table 6: S1-SELECT decisions, run 1. Distinct strings: steps whose five proposals are not all identical. Distinct actions: after merging proposals with the same tools and key clinical arguments. Mean max p and entropy (normalised by $\log n$) describe the System One model's distribution where there was a choice. Departs: share of those steps where it chose a proposal other than the LLM's first sample.

type that earlier executed submissions have not yet matched, with ClinEnv's pair scores (ICD retrieval and re-ranking with hierarchical F1 for diagnoses and procedures; ATC hierarchy with an action gate for medications); a proposal is right if its score is at least 0.5.

Canonical and injected sets. Each proposal is rendered as action plus canonical name (the ICD title of its mapped code, or the ATC-normalised drug name), without the rationale argument, and duplicates are merged. Where no option is right, one wrong option, chosen with a seed, is replaced by an open gold decision rendered the same way; a single wrong option gets the gold one added. States are replayed exactly from an append-only message log.